

Document for

Generalized Open-set Single-shot Character Recognition on Ancient Egyptian Hieratic Characters

Requirement / Test bench spec

CPU: anything with AVX2 / R5 5500

* Intel or AMD, I have no idea if this works on Zhaoxin or smth to that nature

GPU: Nvidia GPU (Pascal or newer) / GTX1070

Ram: 8 Gib/ 32Gib

Disk: Yes / 320Gib HDD

OS: linux / Debian 12.14

Internet: Yes / Yes

Note: this document is not for power users.

This is **NOT** how things should be done,
but rather a quick way to just run it on a testbench.

Audit your code and script and set environments properly,
and please do not blindly run random bash scripts off internet.

Trivia: Maunl, aka Pallas' cat, is the oldest cat specie on earth today.

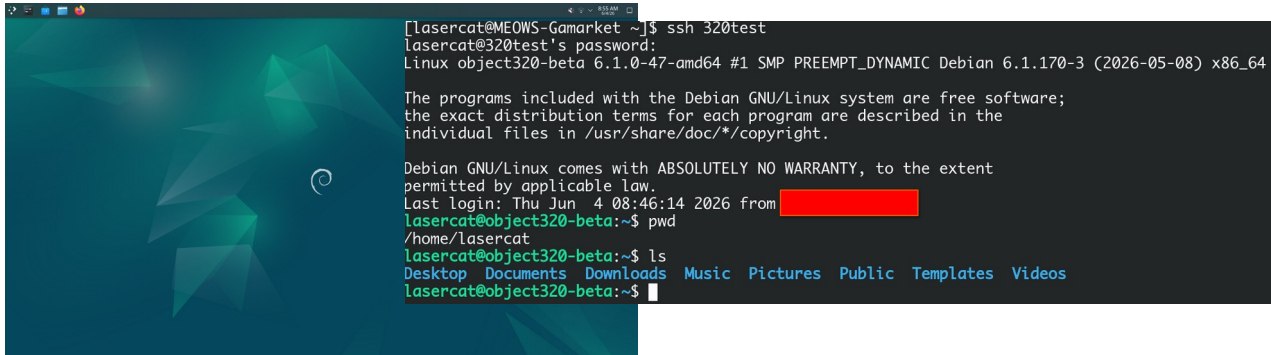


1. Setup Environment

System & Cuda

1. install debian 12.14 KDE, select ssh if you need it, use lasercat as username

you can mod the code later to fit your username and paths...



```
[lasercat@MEOWS-Gamarket ~]$ ssh 320test
lasercat@320test's password:
Linux object320-beta 6.1.0-47-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.170-3 (2026-05-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Jun  4 08:46:14 2026 from [REDACTED]
lasercat@object320-beta:~$ pwd
/home/lasercat
lasercat@object320-beta:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
lasercat@object320-beta:~$
```

2. download script from <https://github.com/lancercat/320-env-debian> to ~/Downloads

3. set up basic environment

su

bash basicenv.sh

4. **reboot** (bcs the kernel version may bump, so this is a **MUST**)

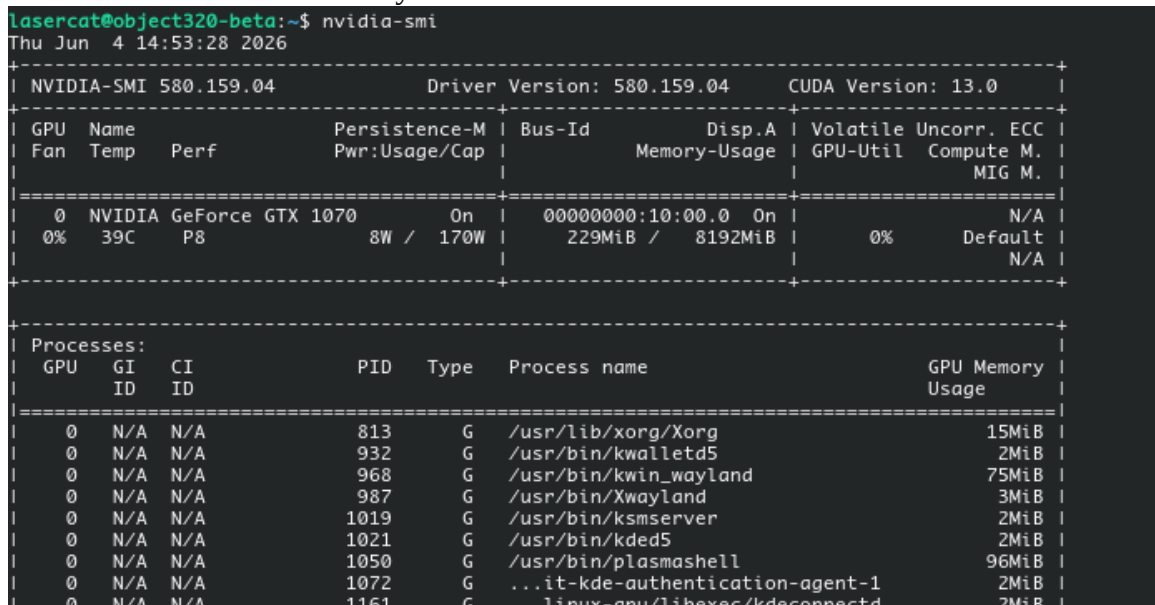
5. install cuda

su

bash setupenv.sh

6. reboot (for nvidia driver to kick in)

7. run nvidia-smi to check that you have it



```
lasercat@object320-beta:~$ nvidia-smi
Thu Jun  4 14:53:28 2026

+-----+
| NVIDIA-SMI 580.159.04                  Driver Version: 580.159.04          CUDA Version: 13.0     |
+-----+-----+
| GPU   Name                               Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC | |
| Fan  Temp  Perf              Pwr:Usage/Cap |      Memory-Usage | GPU-Util  Compute M. |
|               |                    |                      | MIG M. |
+-----+-----+
|  0  NVIDIA GeForce GTX 1070              On | 00000000:10:00.0 On |         N/A |
|  0%   39C   P8               8W / 170W | 229MiB / 8192MiB |      0%   Default |
+-----+-----+

+-----+
| Processes: |
| GPU   GI   CI        PID   Type   Process name                      GPU Memory |
|               ID   ID                 |                 Usage |
+-----+-----+
|  0  N/A  N/A         813     G   /usr/lib/xorg/Xorg                  15MiB |
|  0  N/A  N/A         932     G   /usr/bin/kwalletd5                  2MiB |
|  0  N/A  N/A         968     G   /usr/bin/kwin_wayland               75MiB |
|  0  N/A  N/A         987     G   /usr/bin/Xwayland                   3MiB |
|  0  N/A  N/A        1019     G   /usr/bin/ksmsserver                 2MiB |
|  0  N/A  N/A        1021     G   /usr/bin/kded5                      2MiB |
|  0  N/A  N/A        1050     G   /usr/bin/plasmashell                96MiB |
|  0  N/A  N/A        1072     G   ...it-kde-authentication-agent-1    2MiB |
|  0  N/A  N/A        1161     G   ...linux-gnu/libexec/kdeconnectd    2MiB |
+-----+-----+
```

Torch & Etc

1. install torch and torch_scatter (a custom fork to fix glog)

```
source ~/.bashrc && bash pyenv.sh
```

2. set up code, weight, and models.

a. prepare dir:

```
sh makdirs.sh
```

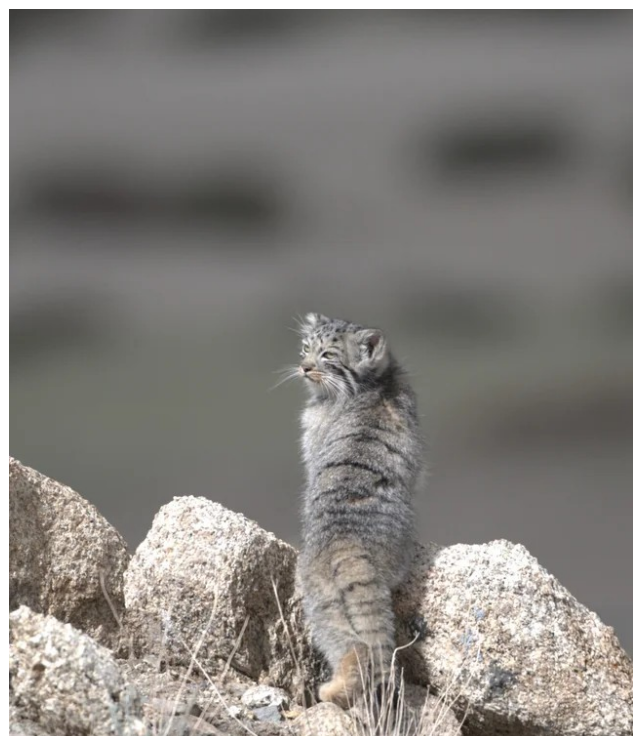
b. clone code

```
cd ~/cat; git clone git@github.com:lancercat/object-320-beta.git
```

c. download data and weight to Downloads, put weights to ~/hydra_saves and data to ~/ssddata

3. check that you have everything

```
lasercat@object320-beta:~$ ls ~/cat
object-320-beta
lasercat@object320-beta:~$ ls ~/ssddata/
dicts_v3  ecr-v3R
lasercat@object320-beta:~$ ls ~/hydra_saves/
b1nf-1024-ecrv3R-abiaug-48-48  b1nf-1024-ecrv3R-noaug-48-48  b1nf-1024-ecrv3RTW-abiaug-48-48
b1nf-1024-ecrv3R-abiaug-48-48-run2  b1nf-1024-ecrv3R-noaug-48-48-run2  b1nf-1024-ecrv3RTW-abiaug-48-48-run2
```



2. Reproducing the results

1. limit your card if you have a shoddy psu

su

nvidia-smi -pm 1 && nvidia-smi -pl 100

2. active your torch

source ~/Downloads/env/pytorch_env/bin/activate

3. go to eval script dir

cd ~/cat/object-320-beta/neko_2025_NGNW/beta320

4. run inference

PYTHONPATH=\${PWD}/../../../../../ python eval_batch.py | tee hee.log

if you see this, its running,

```
INFO: let me remind you that queue contents can not be restored for now
INFO: Module per_inst_cls_loss dose not support saving
Date: 2026-06-04-15-51-11 ,TEST: ecr-v3R_val_fsl_1shot ,Epoch: 0 ,Iter: 80000 ,Total-K: 4151.0 ,ACR: 0.4801252710190316 ,FPS: 369.565444692
22155
Date: 2026-06-04-15-51-22 ,TEST: ecr-v3R_val_osr_1shot ,Epoch: 0 ,Iter: 80000 ,Total-K: 3650.0 ,ACR: 0.6526027397260274 ,FPS: 369.390240499
52695 ,Recall: 0.249500998003992 ,Precision: 0.2997601918465228 ,FScore: 0.27233115468409586 ,Total-U: 501.0
```

utilization btw, note we are running off hdd so the io is not ideal.

```
Device 0 [NVIDIA GeForce GTX 1070] PCIe GEN 3@16x RX: 716.0 MiB/s TX: 6.532 MiB/s
GPU 1708MHz MEM 3802MHz TEMP 55°C FAN 0% POW 99 / 100 W
GPU[ ] 63% MEM[ ] 17.017GiB/8.000GiB
```

5. extrat performance

awk '/% ====/ {flag=1; next} flag' hee.log > reproduce.

tex

5. compare it to the paper report value

Note there could be minor performance differences moving from environment to environment

```
(pytorch_env) lasercat@object320-beta:~/cat/object-320-beta/neko_2025_NGNW/beta320$ diff reproduce.tex ../../documents/reproduce.txt
0a1
> % =====
1a3
> % =====
4c6
< \newcommand{\NoursFs1ValAcrAvg}{45.43}
---
> \newcommand{\NoursFs1ValAcrAvg}{45.44}
12c14
< \newcommand{\NoursOsrValAcrAvg}{64.83}
---
> \newcommand{\NoursOsrValAcrAvg}{64.82}
16c18
< \newcommand{\NoursOsrValPrecisionAvg}{29.54}
---
> \newcommand{\NoursOsrValPrecisionAvg}{29.52}
18,19c20,21
```



3. Train a model

1. limit your card if your psu came from a pawn shop or smth

su

nvidia-smi -pm 1 && nvidia-smi -pl 100

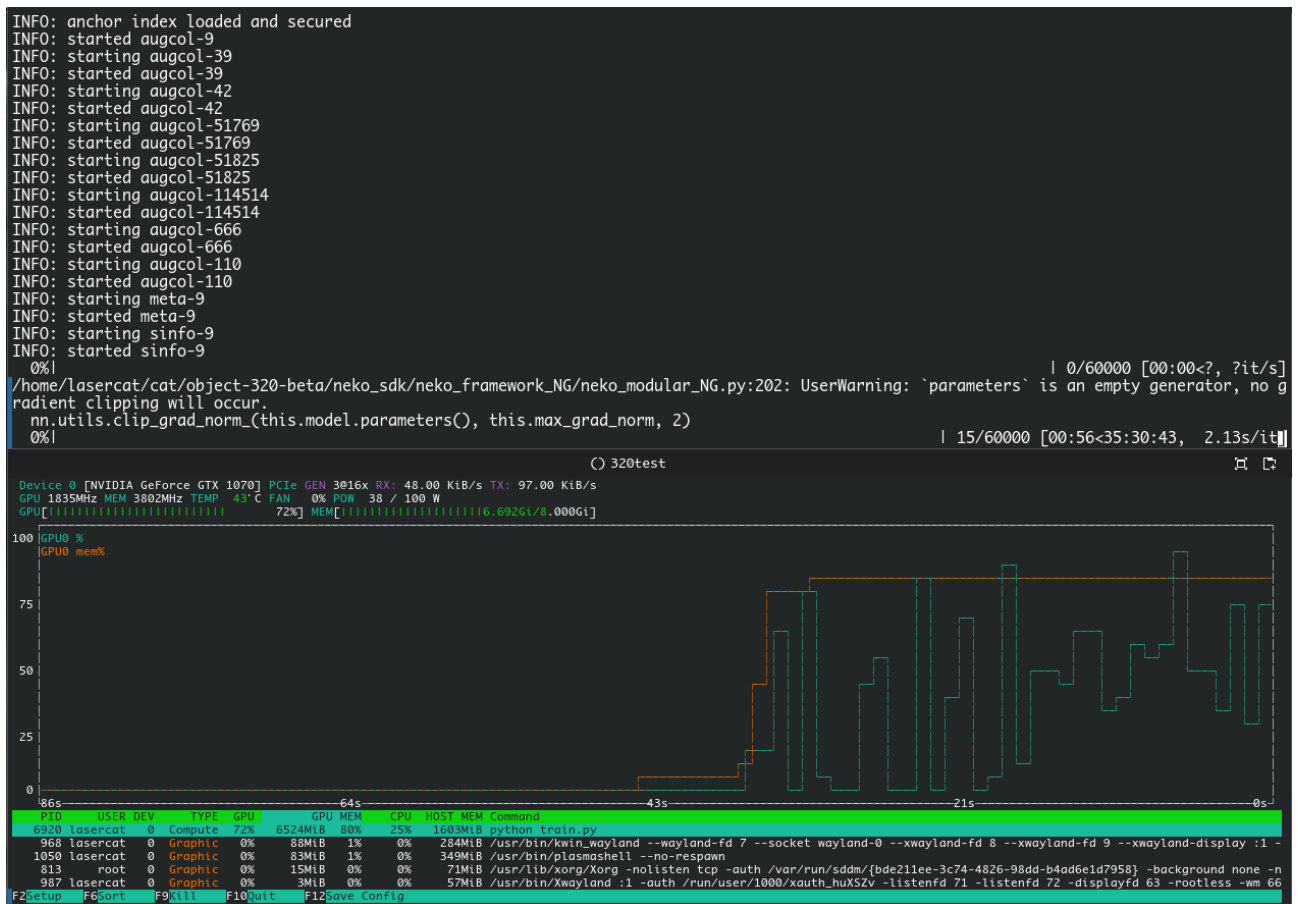
2. go to a model dir, say b1nf-1024-ecrv3R-noaug-48-48

cd ~/cat/object-320-beta/neko_2025_NGNW/beta320/b1nf-1024-ecrv3R-noaug-48-48

3. train

PYTHONPATH=\${PWD}/../../../../../ python train.py | tee hee.log

If you see this, you are training. Utilization btw



That's all for now, GLHF

